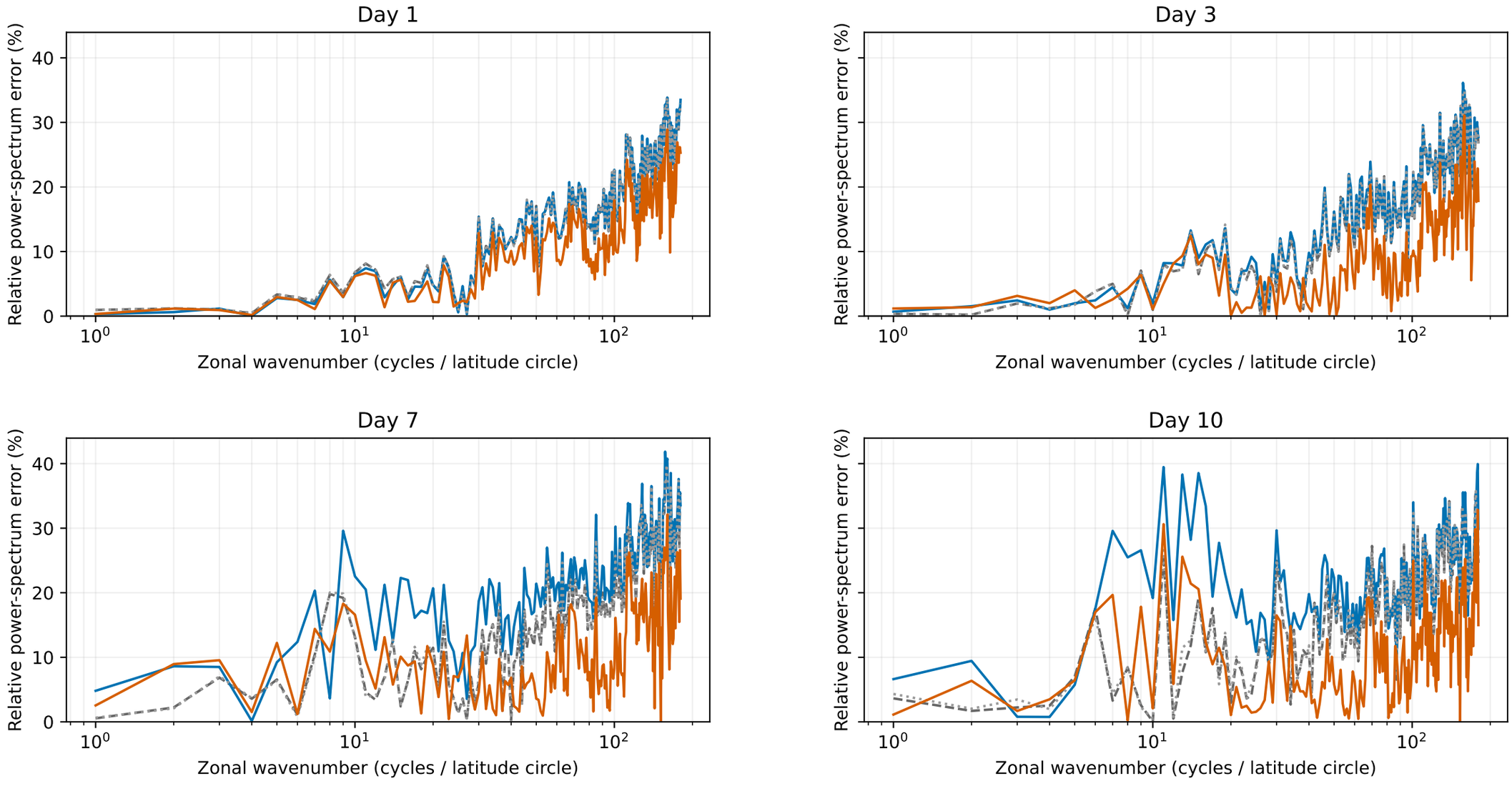
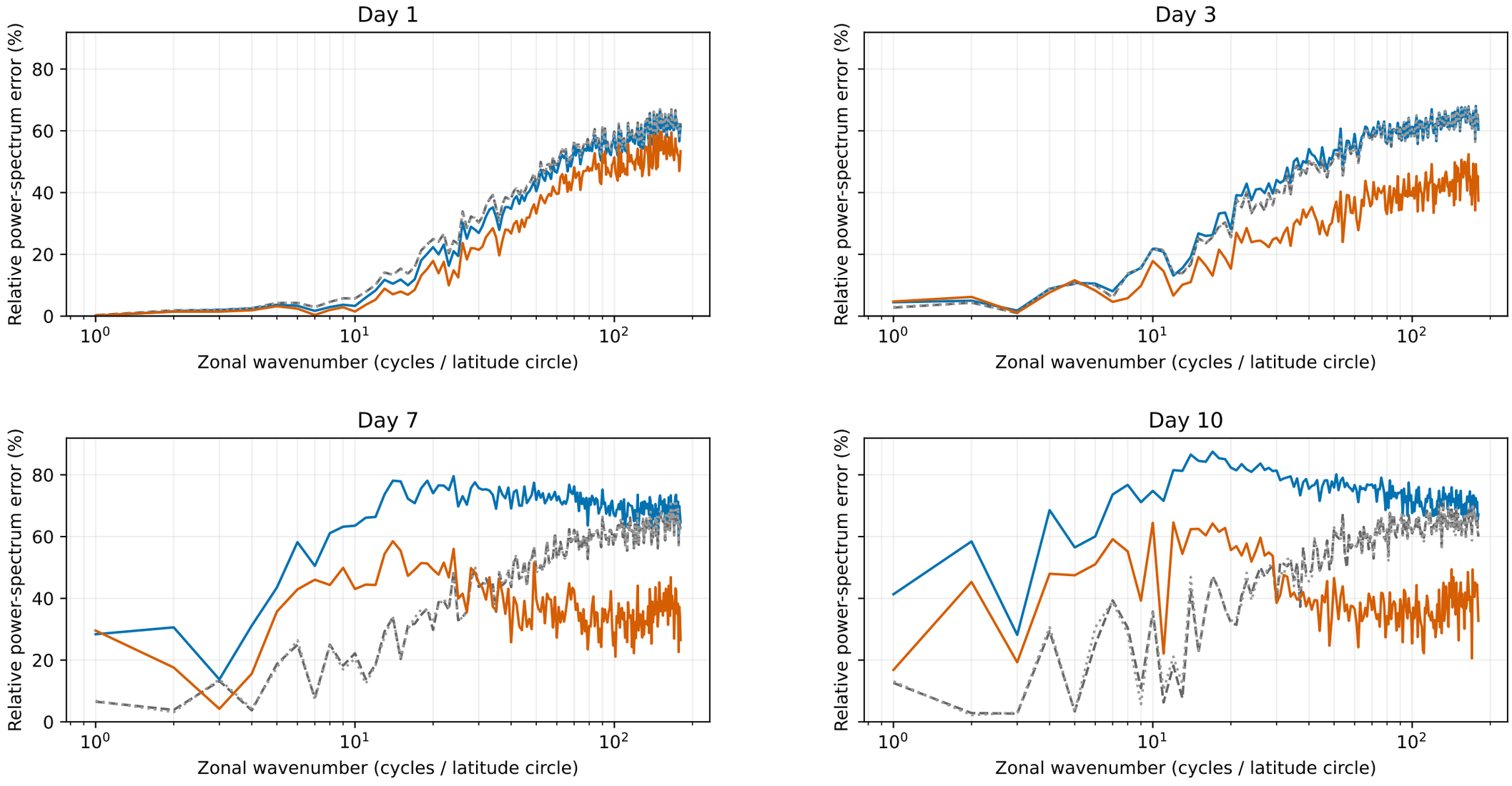
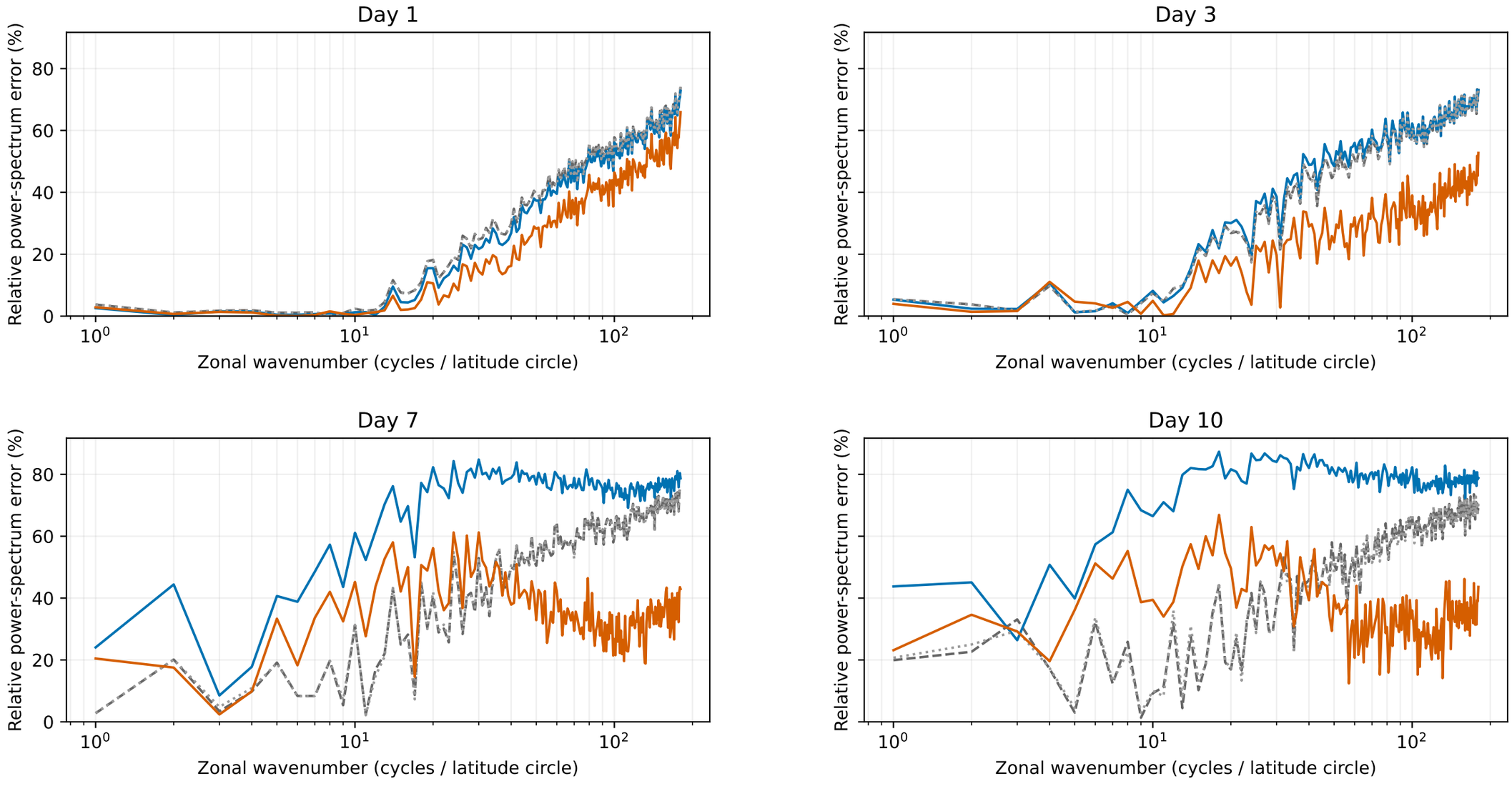

T2m: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


U10: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

V10: relative power-spectrum error (%) · step2000

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better

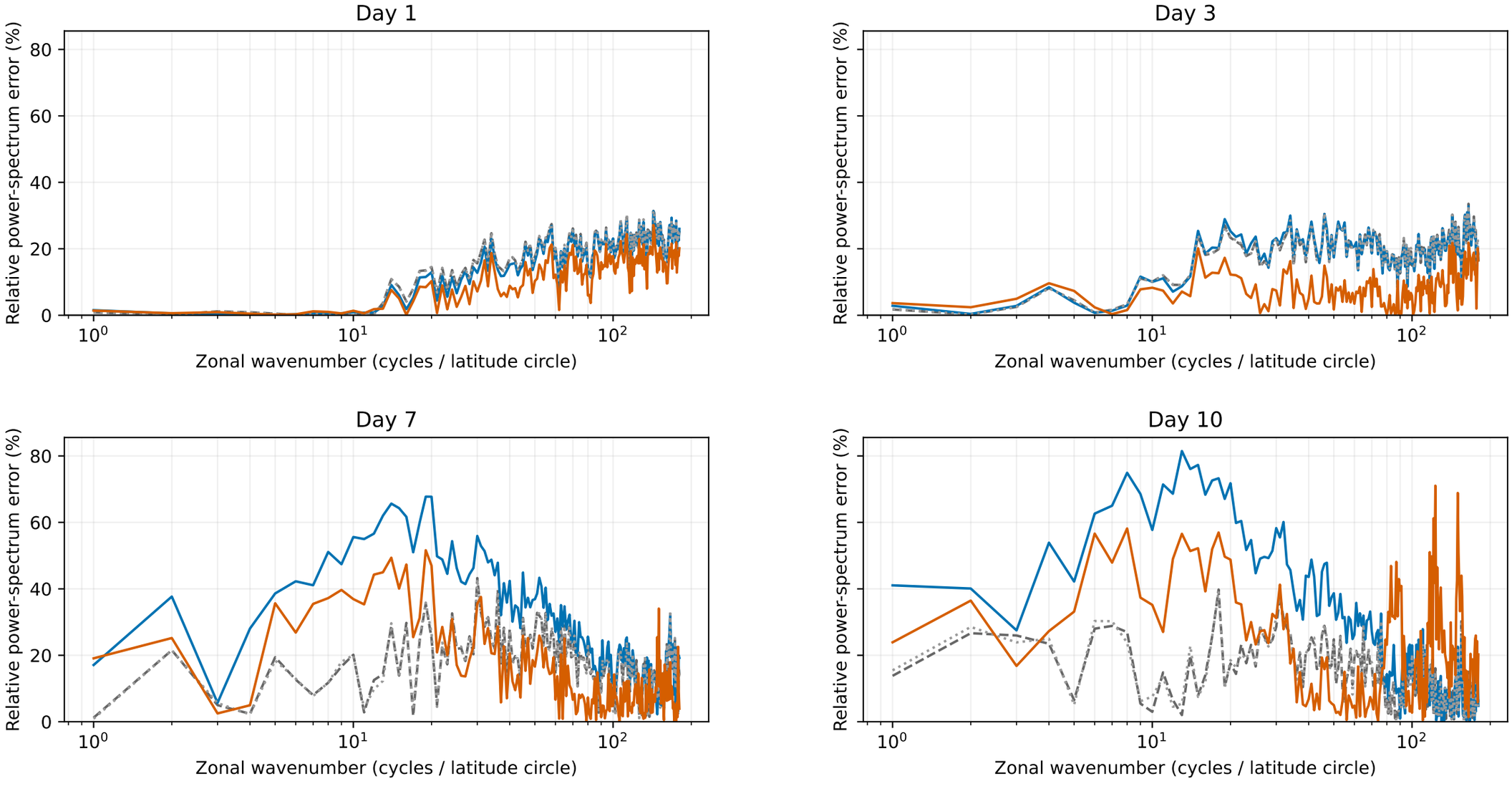


1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios

Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

MSLP: relative power-spectrum error (%) · step2000

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better

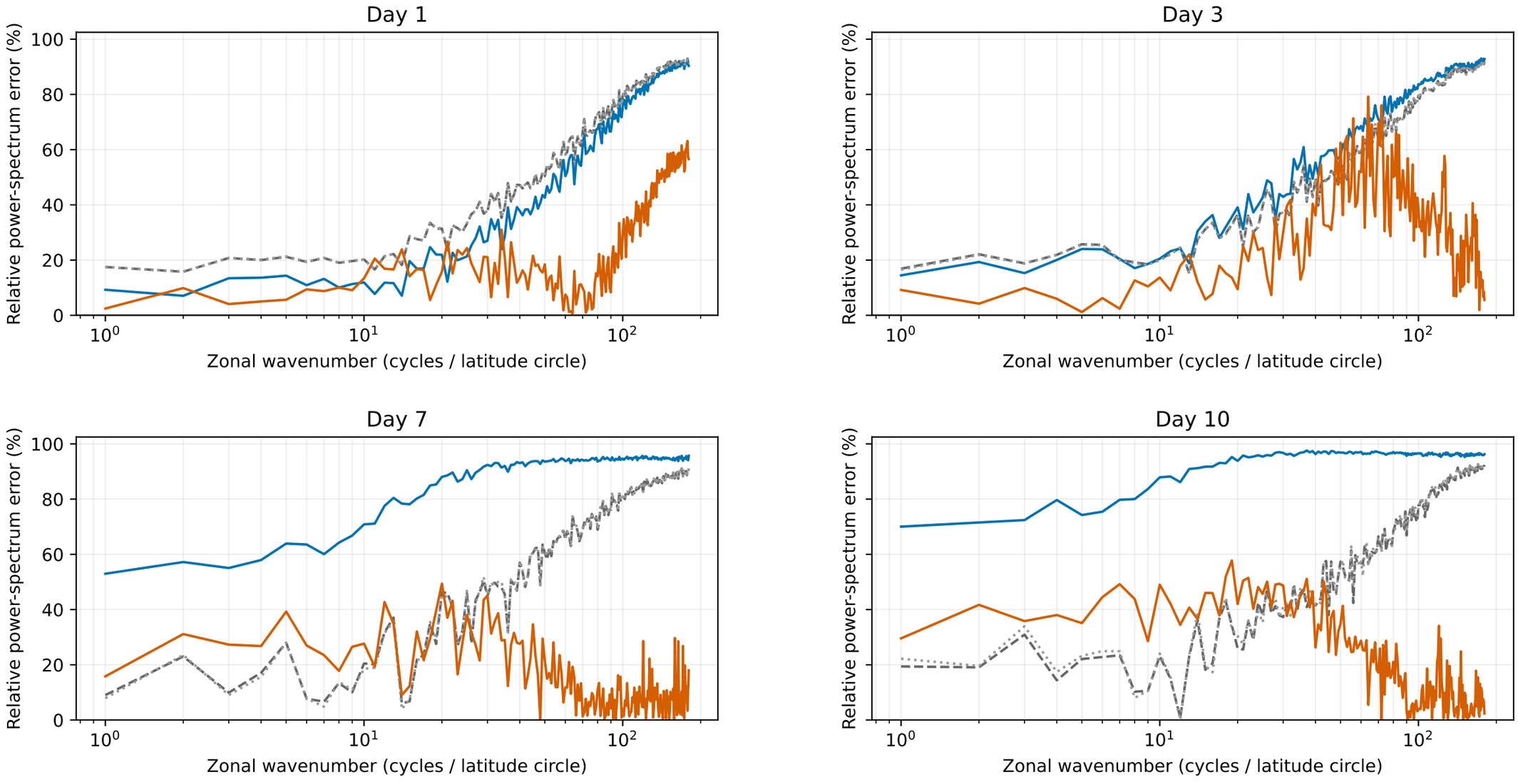


1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios

Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

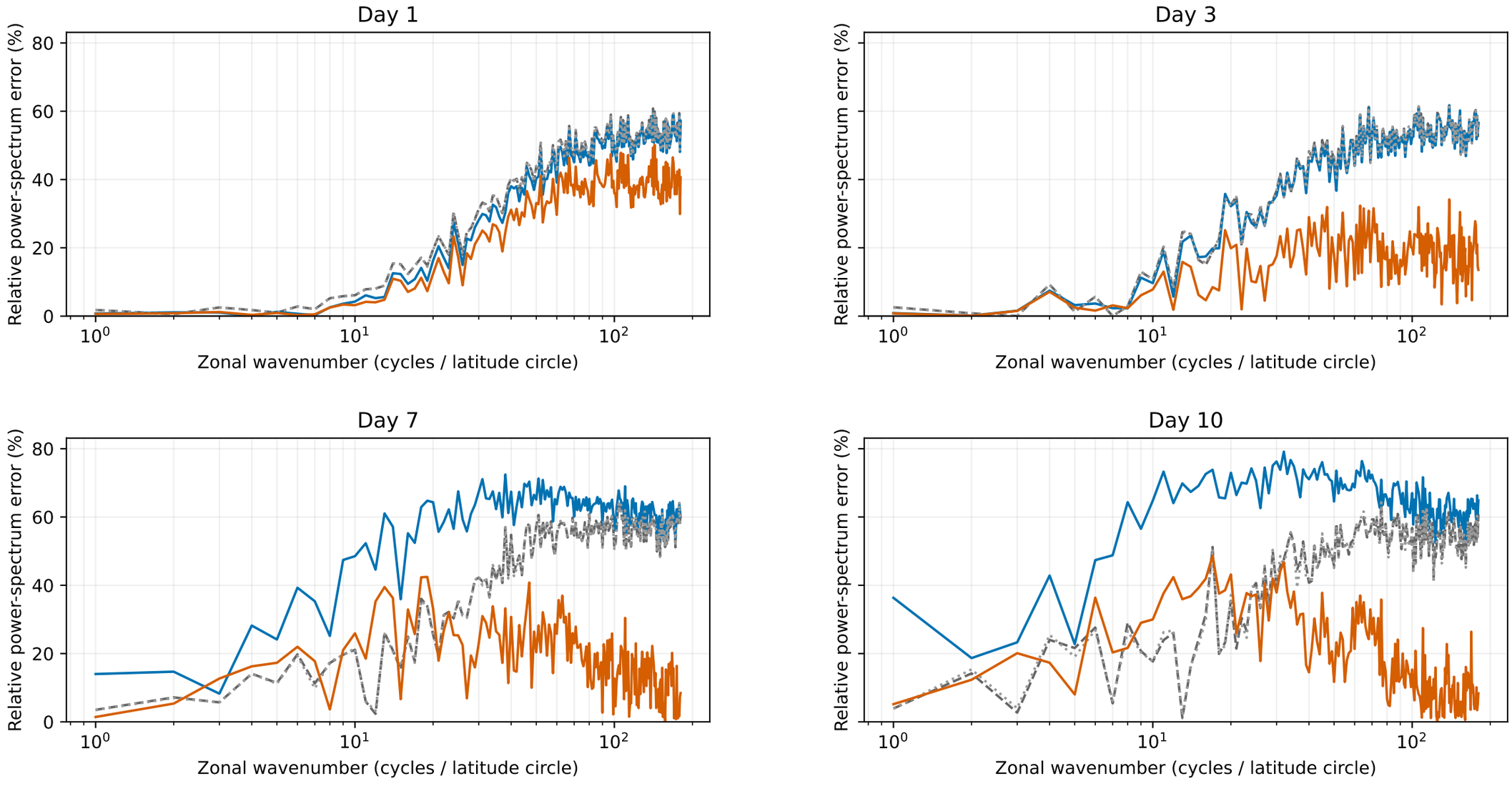
TP6: relative power-spectrum error (%) · step2000

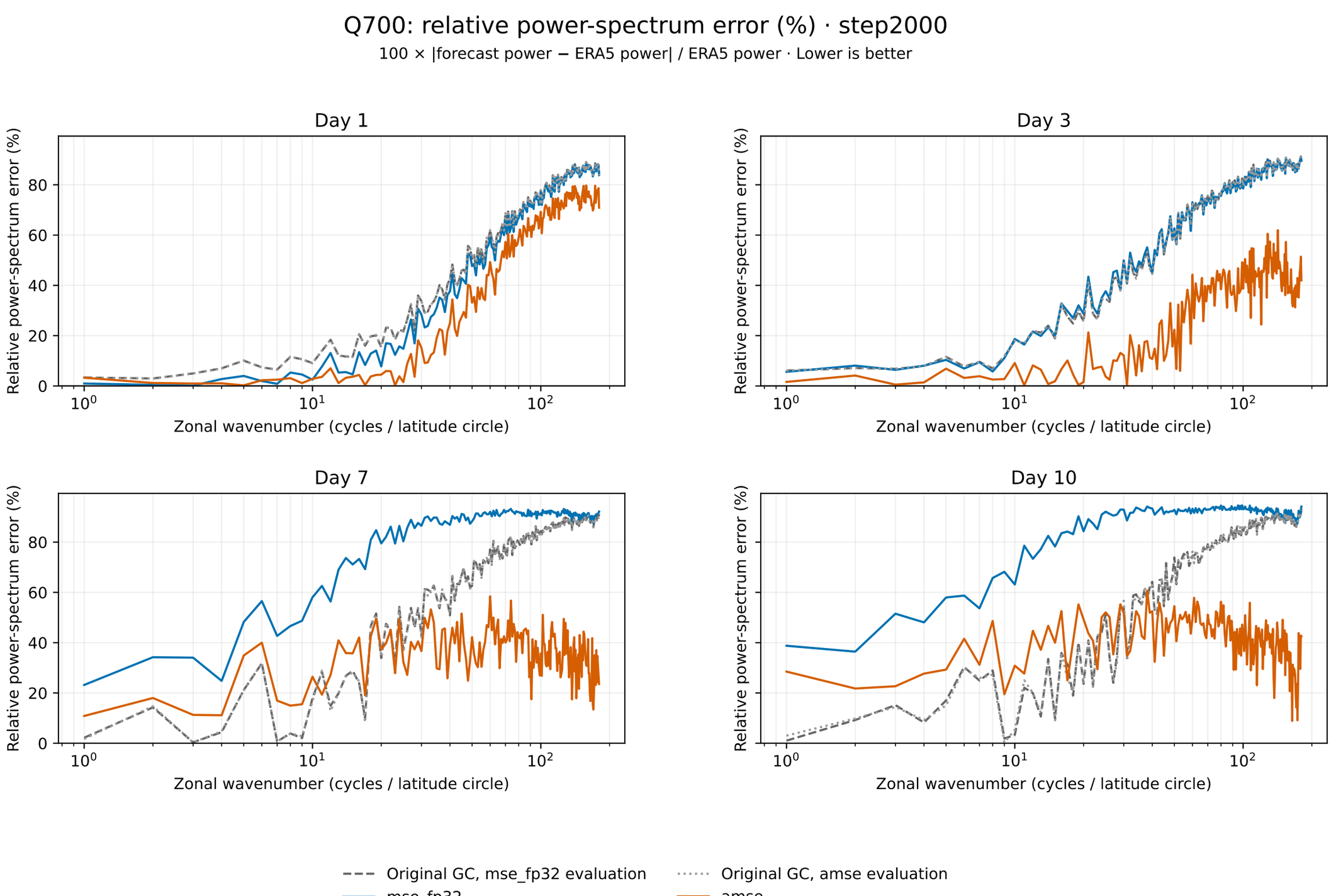
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better

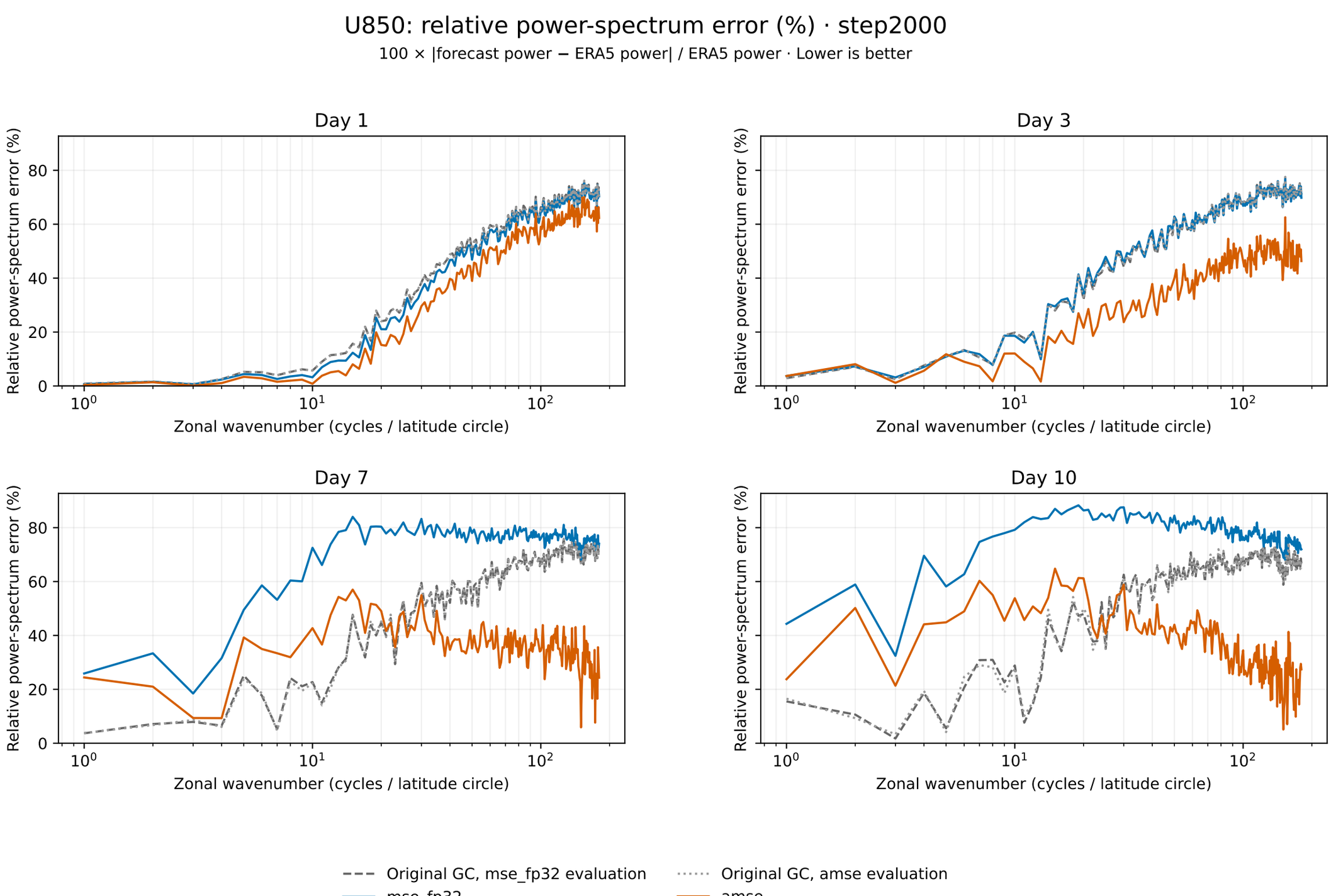


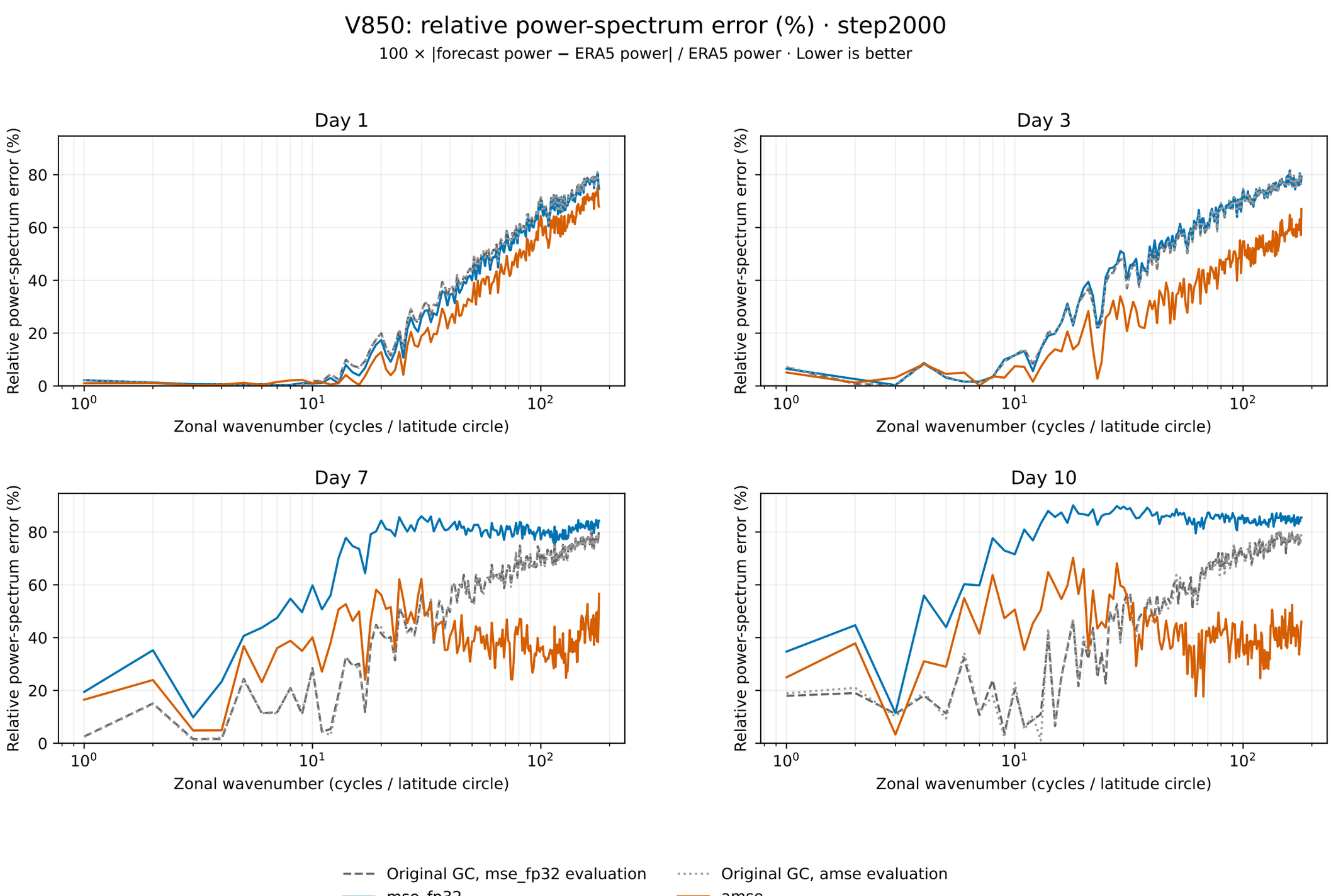
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios

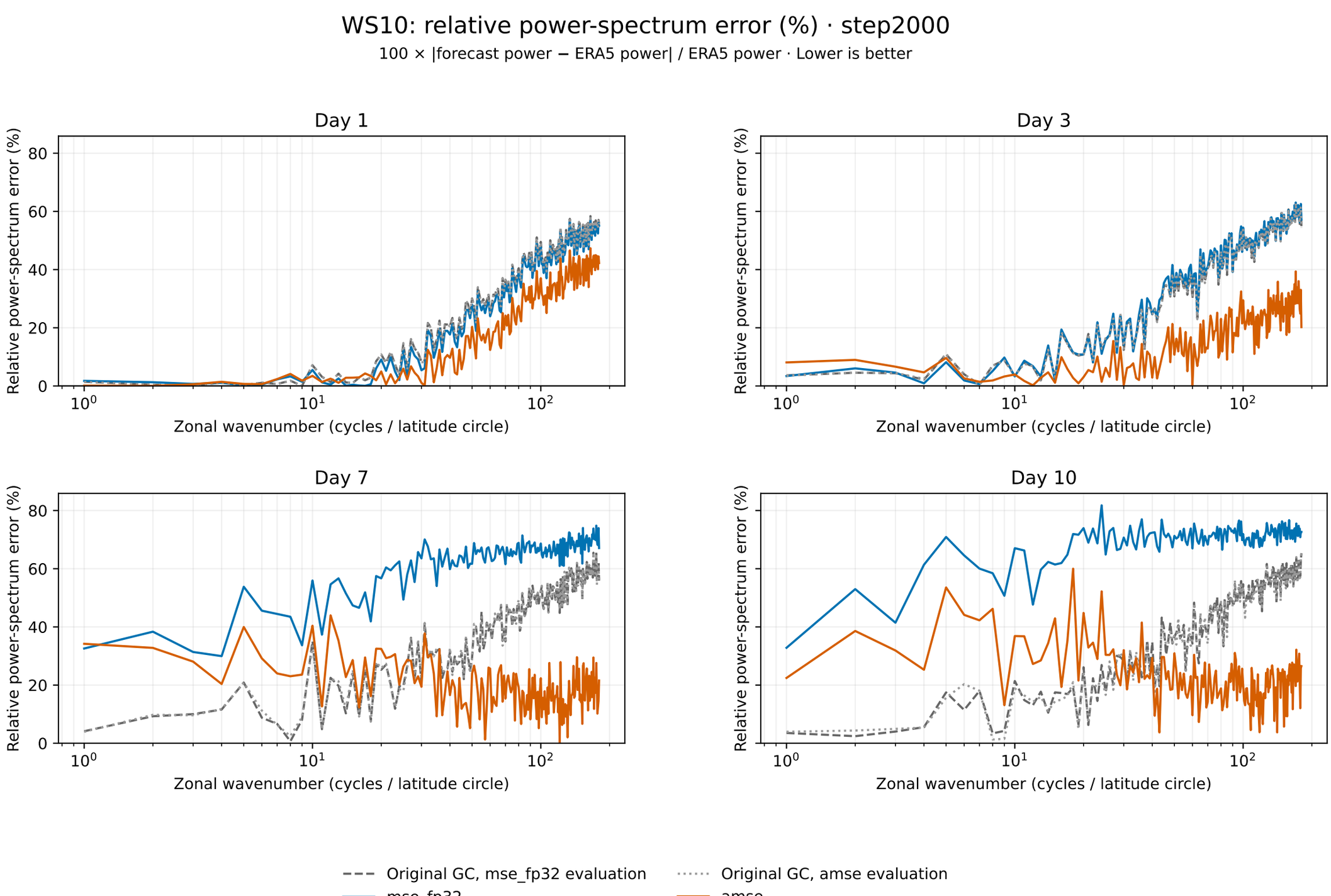
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

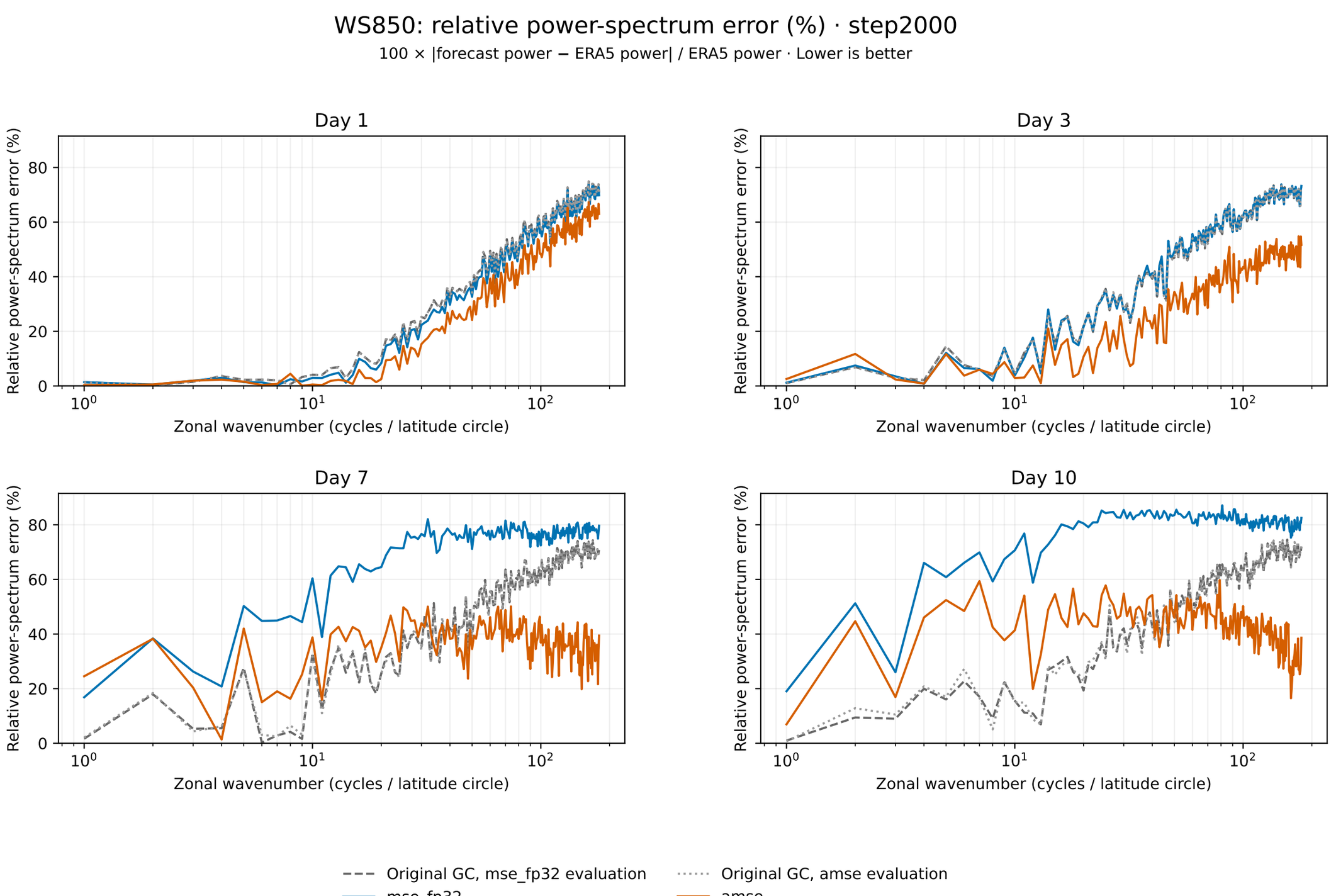

T850: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

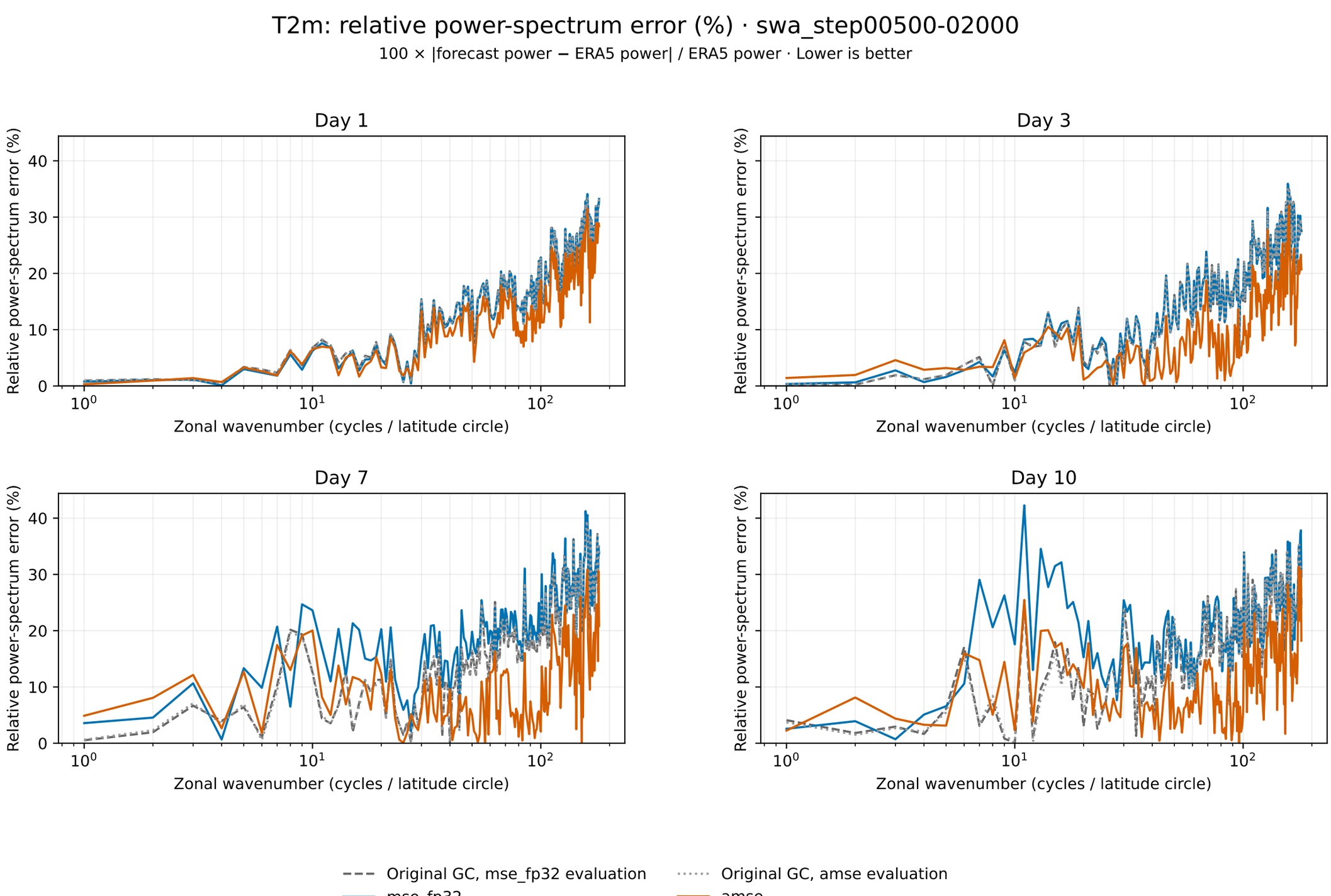

Q700: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

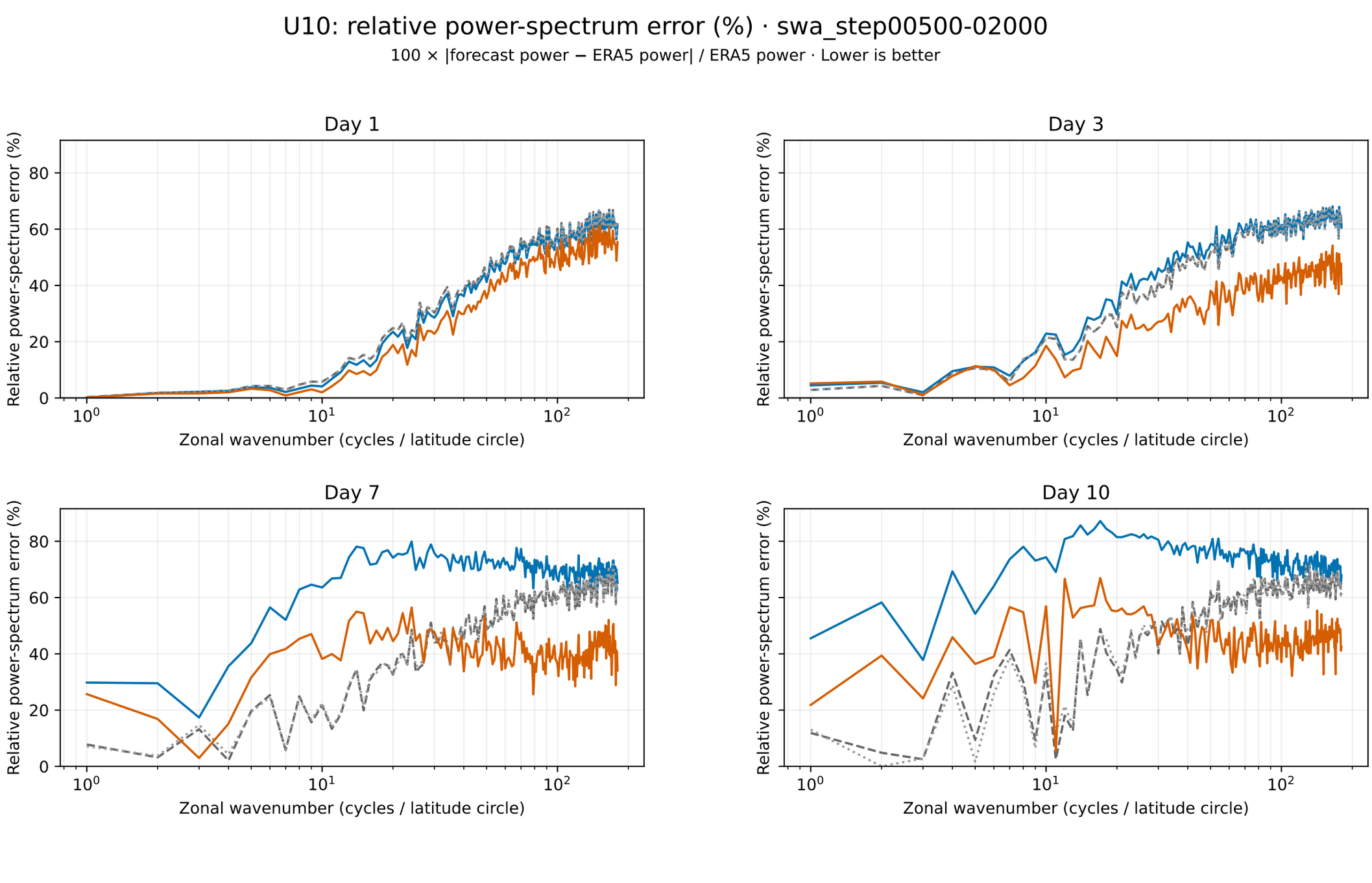

U850: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

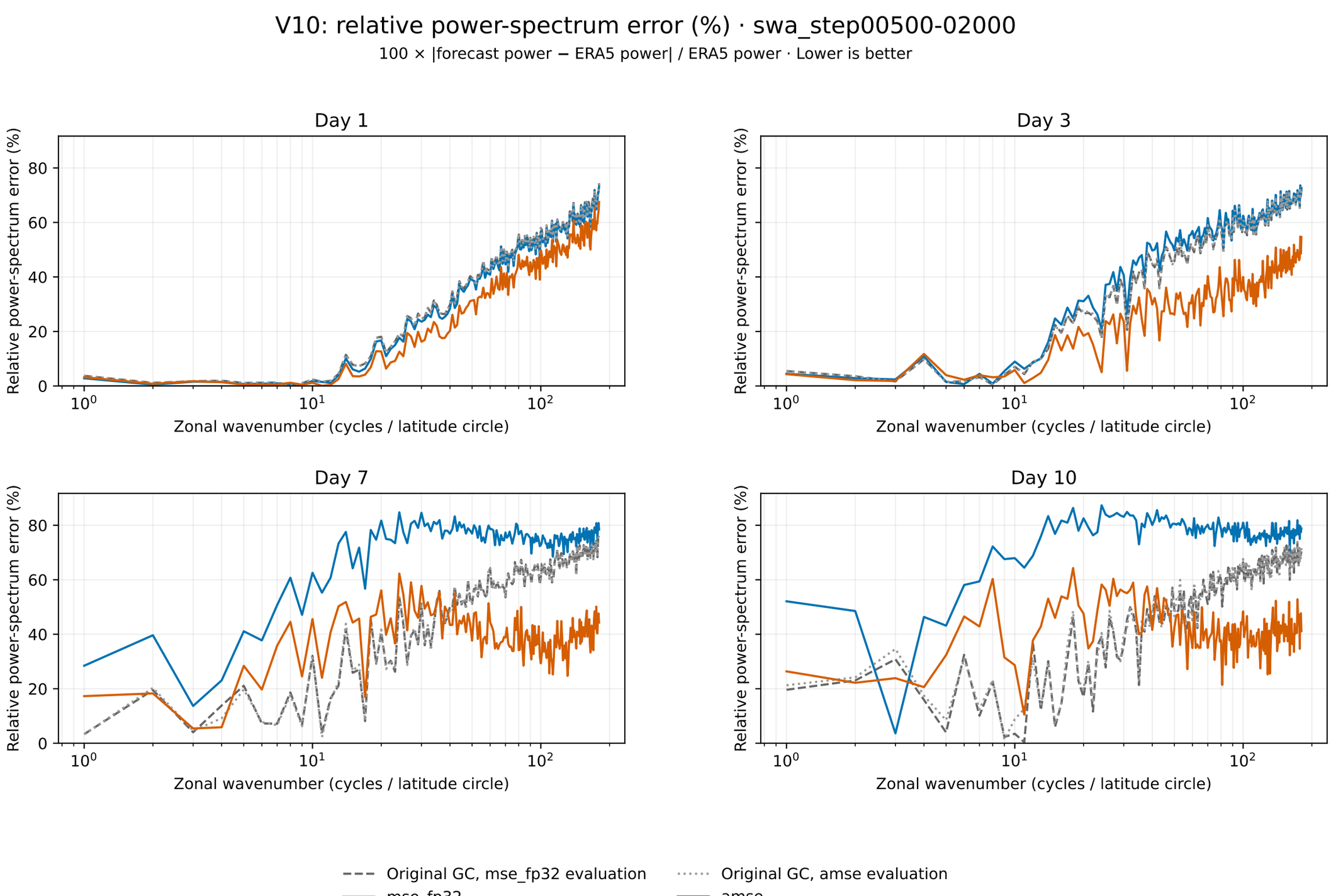

V850: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


WS10: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


WS850: relative power-spectrum error (%) · step2000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

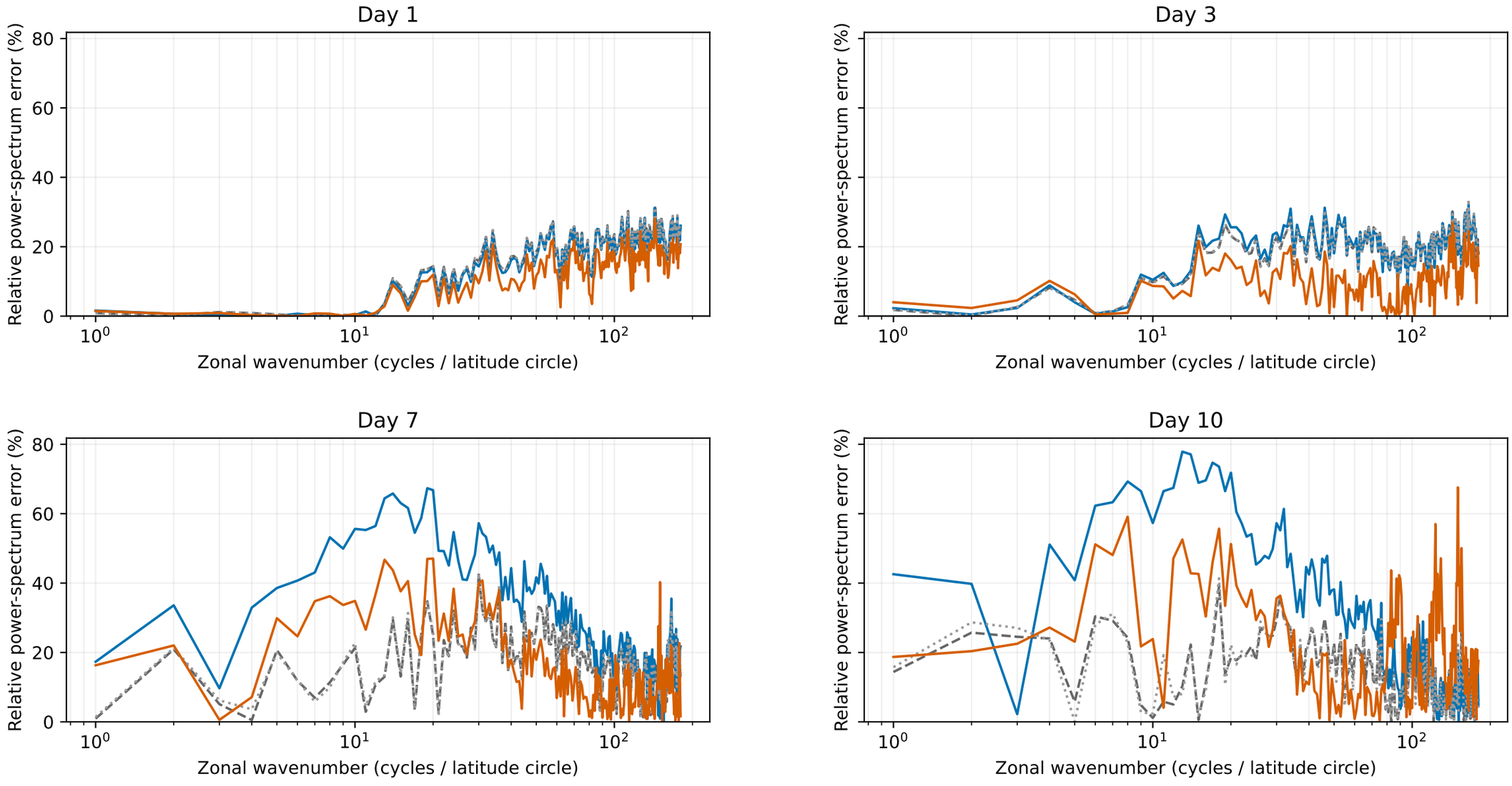

T2m: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


U10: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


V10: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

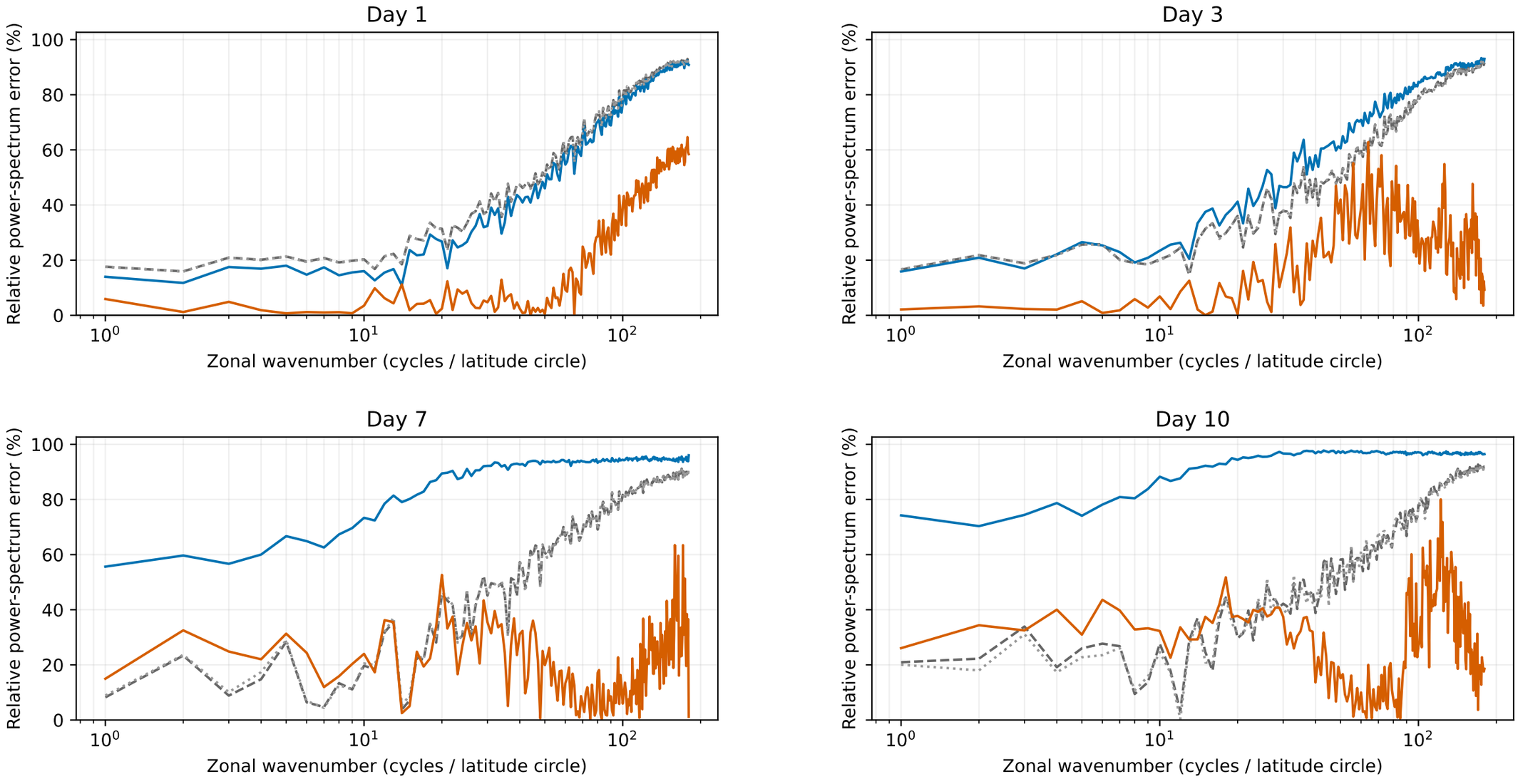
MSLP: relative power-spectrum error (%) · swa_step00500-02000

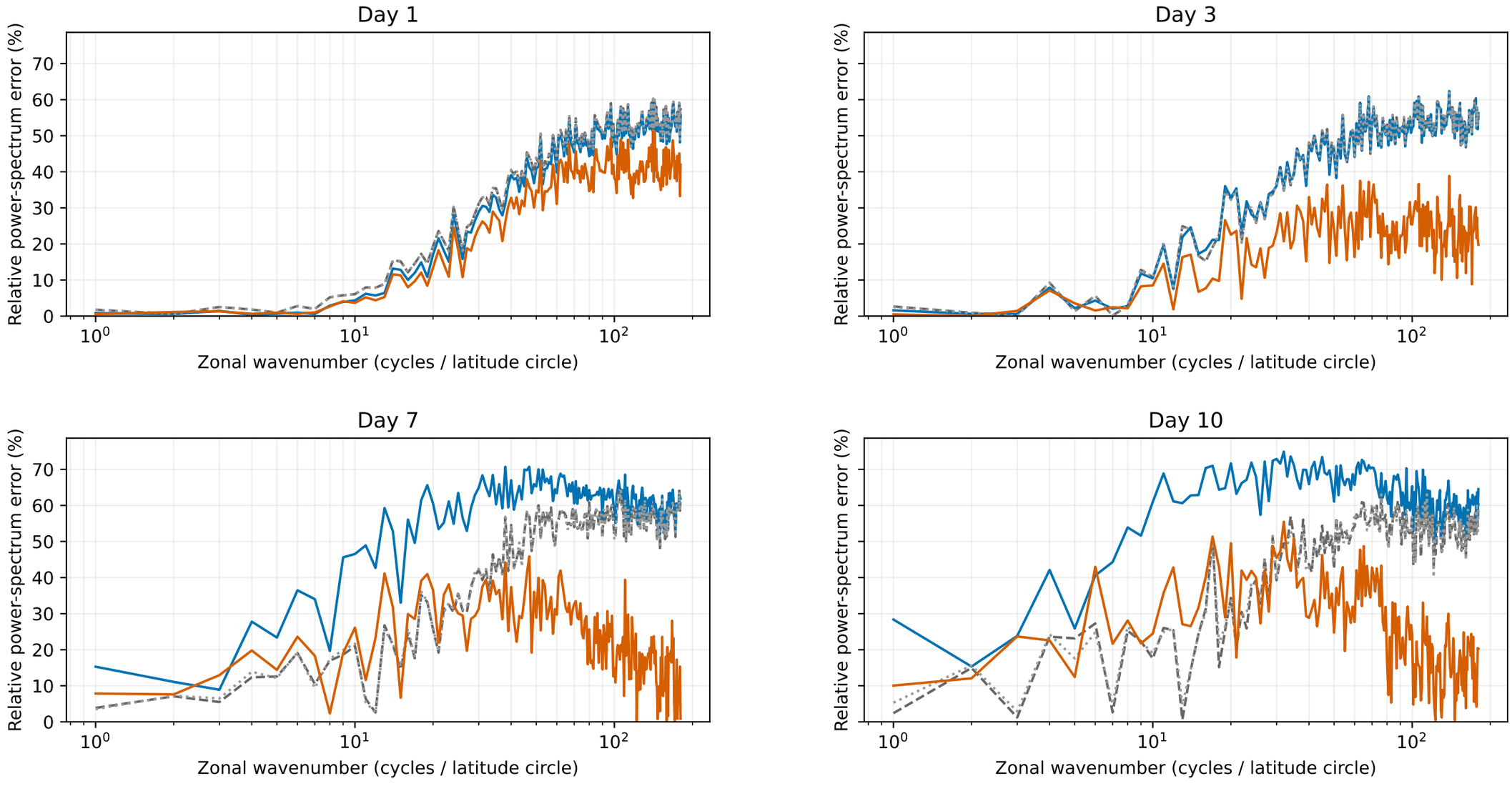
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better

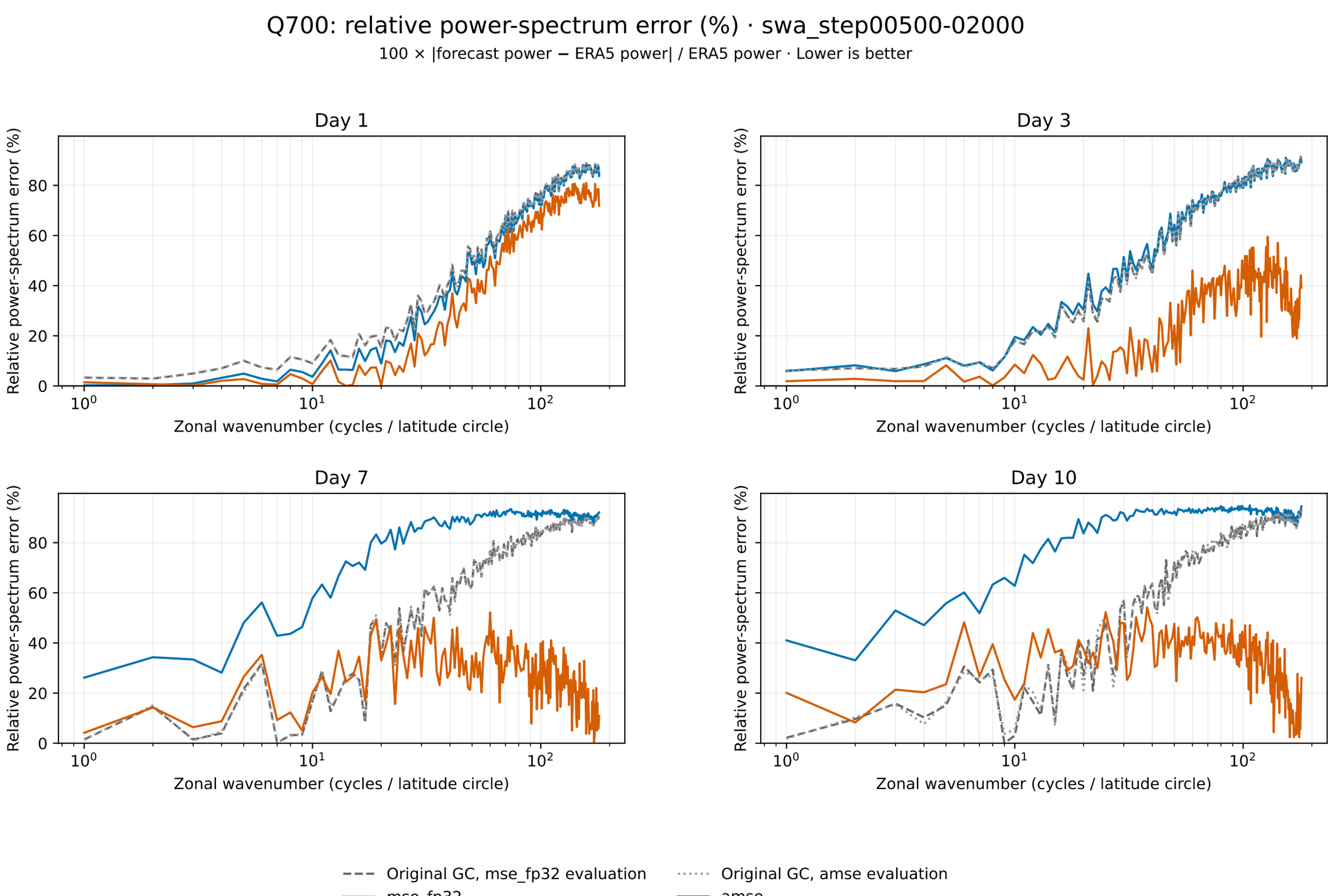


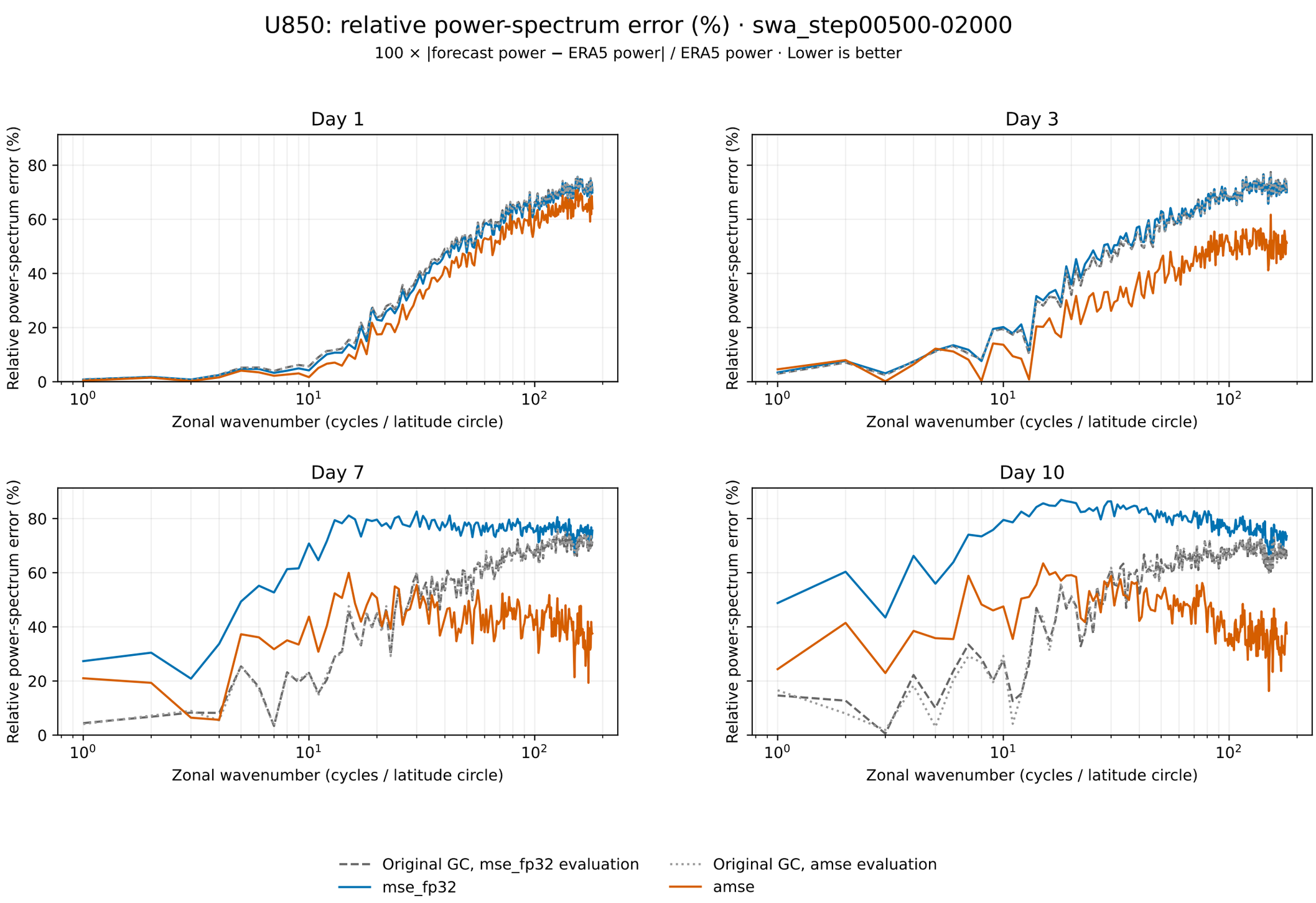
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios

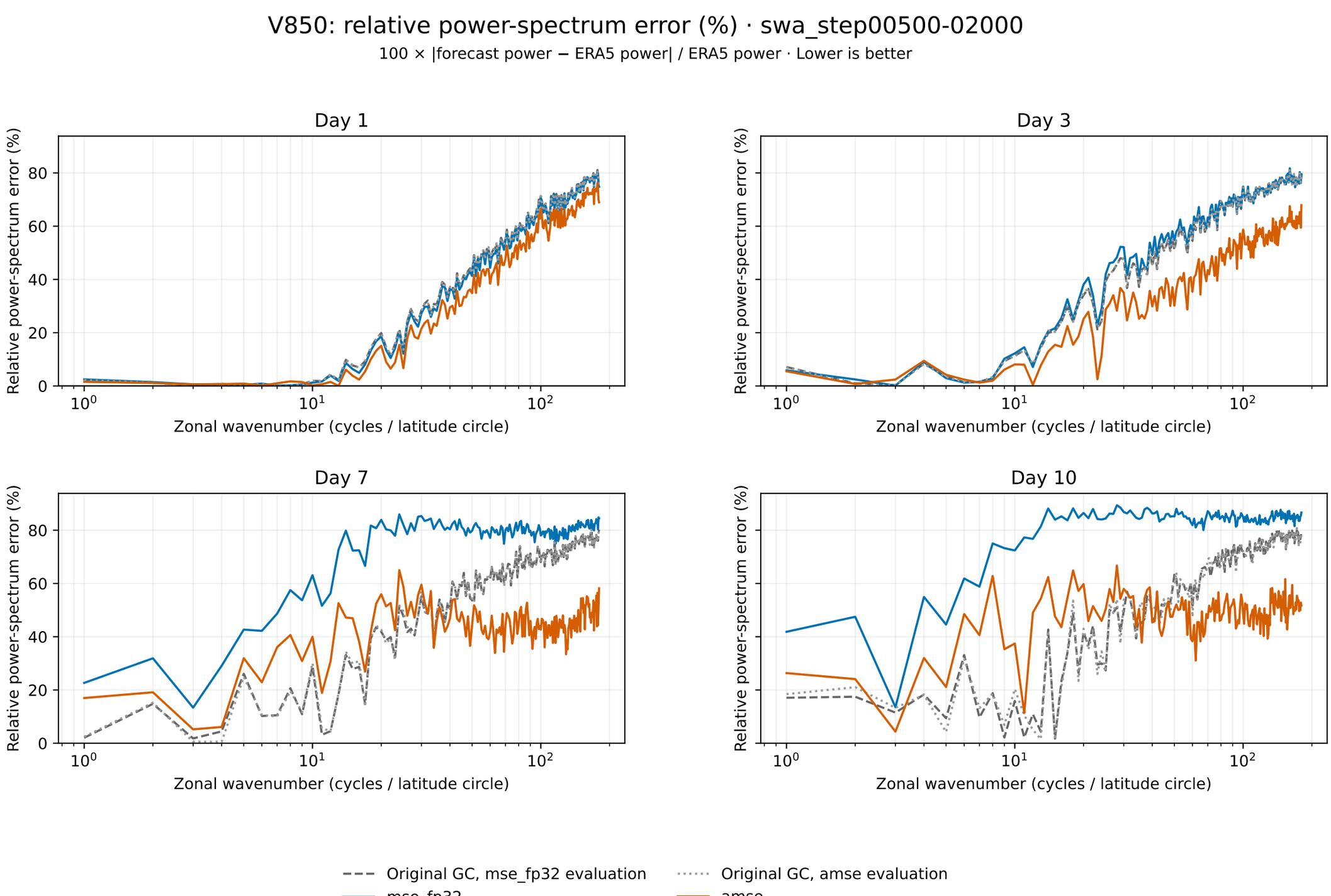
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

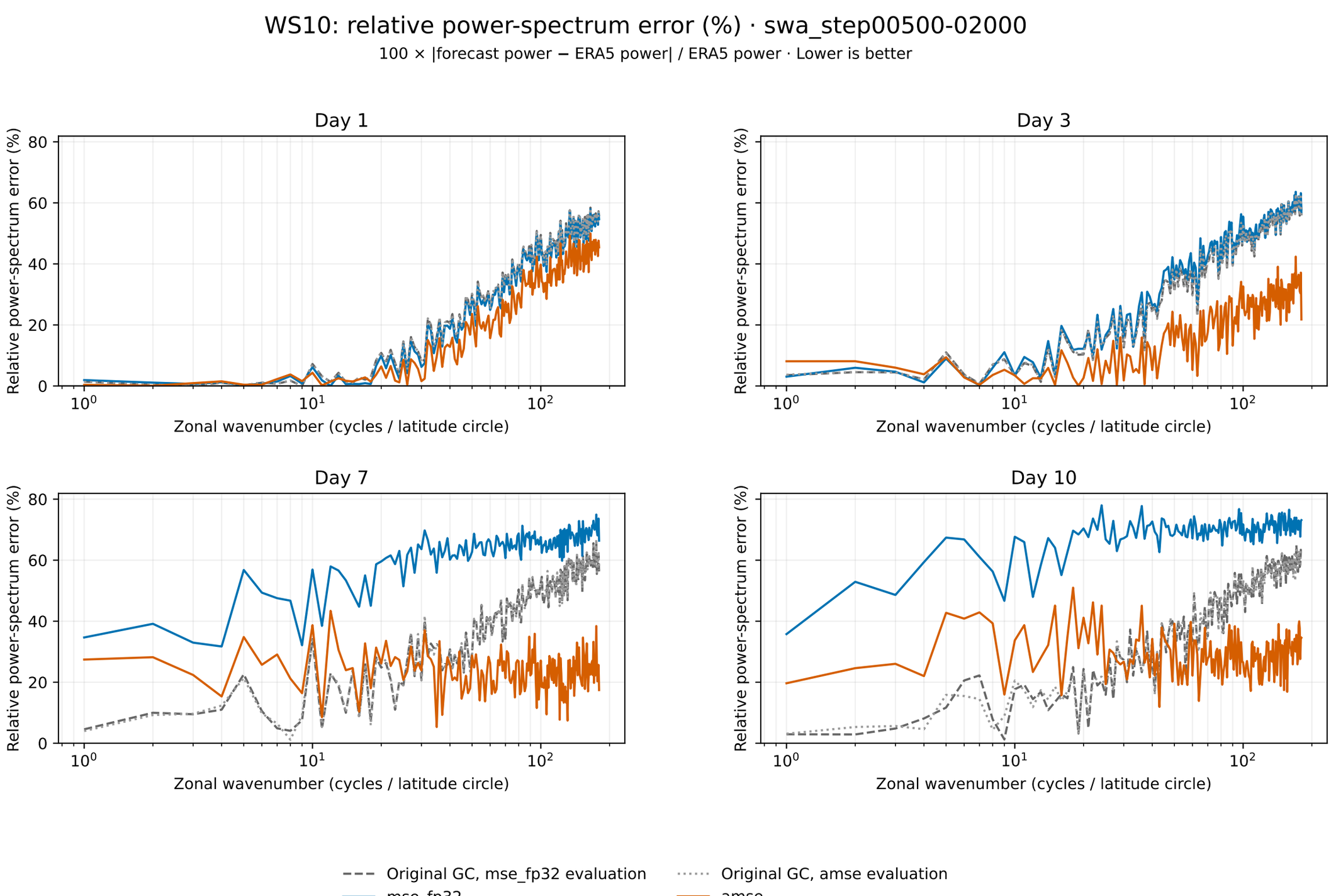

TP6: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.

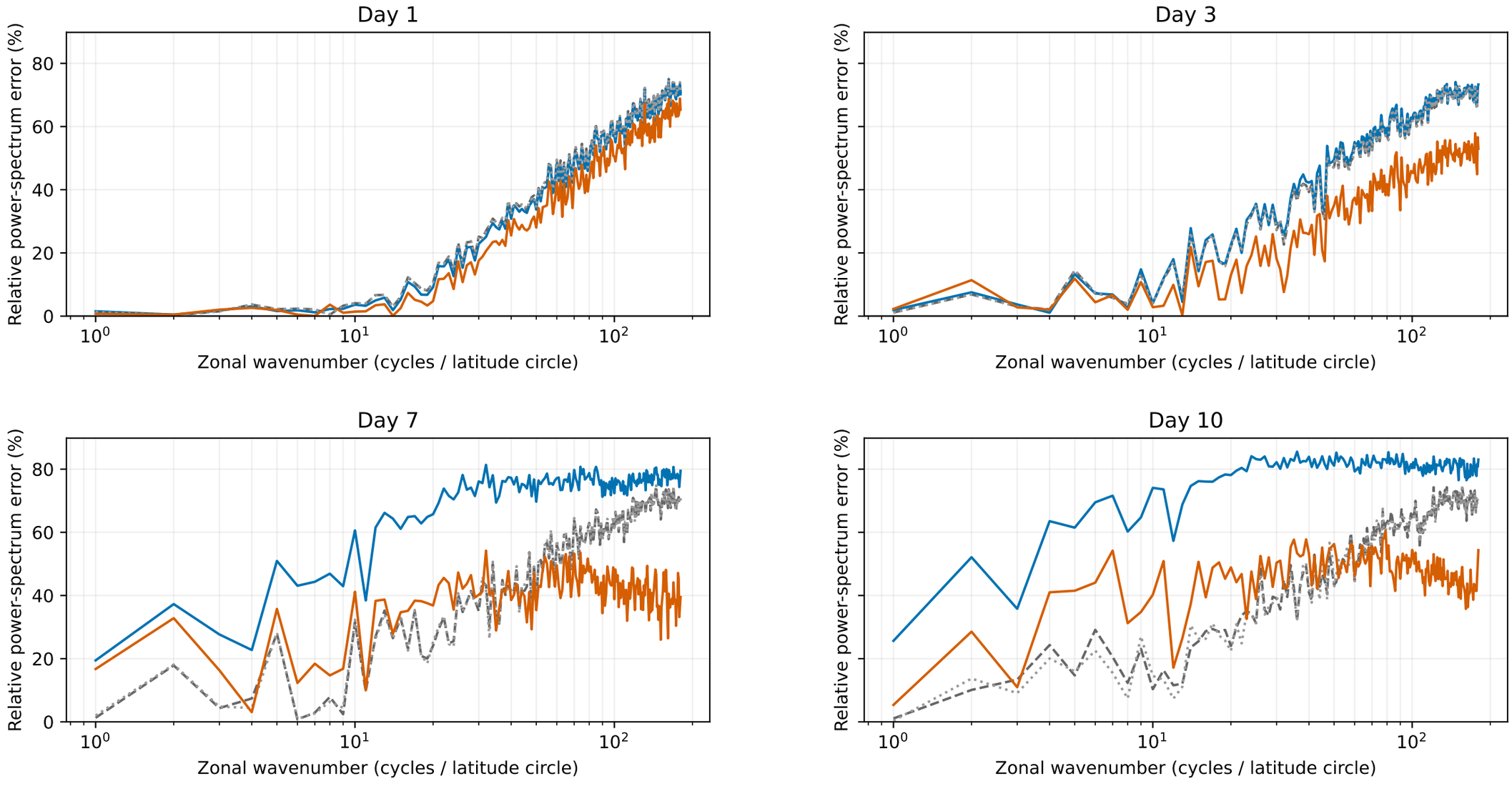

T850: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


Q700: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


U850: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


V850: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


WS10: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
Original GC, amse evaluation
mse_fp32
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.


WS850: relative power-spectrum error (%) · swa_step00500-02000
100 × |forecast power − ERA5 power| / ERA5 power · Lower is better
Day 1
Day 3
Day 7
Day 10
Relative power-spectrum error (%)
Zonal wavenumber (cycles / latitude circle)
Original GC, mse_fp32 evaluation
mse_fp32
Original GC, amse evaluation
amse
1° · Eight fixed 2023 starts · 30–60° both hemispheres · Pooled powers before ratios
Repeated GC baselines differ beyond tolerance; each paired baseline is shown.